Seminar 4 - Lectures 7, 8 and 9
Computer Vision 1, Master AI, 2026

Exercises inherited from Arnoud Visser, Martin Oswald, Qi Bi, Roan van Blanken

1 Exercise 1: Image Stitching

Q.1.a What is image stitching?

Q.1.b Describe the basic steps to stitch two images with the same optical center. Try to list some typical algorithms for each step.

Q.1.c Describe the basic steps of RANSAC.

Q.1.d Assume we are using RANSAC to fit a plane in 3D space. Given a set of ten inlier points belonging to a plane, denoted as P_1, P_2, ..., P_{10}, and an additional three outlier points, denoted as O_1, O_2, and O_3. Also, assume that any three points are not collinear. How many points should be sampled each time to define a plane using RANSAC?

Q.1.e Continuing from the previous question, suppose we repeatedly sample points using RANSAC until we successfully find a plane that contains all ten inlier points $(P_1, P_2, ..., P_{10})$. In the worst-case scenario, how many times do we need to sample to ensure this outcome? (Hint: Use the binomial coefficient formula $C_n^k = \frac{n!}{k!(n-k)!}$ to calculate the number of ways to choose k points from n total points.)

Q1.f Assuming the error boundary δ is set to 0 in the third step of the RANSAC algorithm, what is the minimum fraction of points that need to be inliers to correctly define the plane in this scenario?

2 Exercise 2: Bag of Visual Words

Q.2.a Describe the basic steps of the Bag of Visual Words model?

Q.2.b Explain how k-Means clustering can be used to obtain a visual vocabulary.

Q.2.c Briefly summarize the major steps involved in using k-Means clustering to cluster the SIFT features into k words in BoW.

Q.2.d Explain the difference between dense sampling and interest point sampling. Now, consider an image retrieval system with a dataset and 100 query images, evaluated using precision@10. If we use a BoW representation with 10K words and SIFT descriptors, why might BoW with dense sampled patches perform better on a 1,000-image dataset, but interest point sampling performs better on a 10M-image dataset?

Q.2.e What are the weaknesses of the Bag of Visual Words model?

3 Exercise 3: Image Retrieval

Q.3.a Assume there is an image retrieval system. We find 4 true positive (TP) samples and 96 true negative (TN) samples. There is 1 false positive (FP) sample and 2 false negative (FN) samples. Calculate the recall, precision, and accuracy (in percent). Based on these outcomes, explain why accuracy is not a good metric.

Q.3.b Compute the Average Precision (in percent) for the following relevance ranking: $[R, R, N, N, R, R, N, N]$, where R denotes relevant and N denotes not relevant.

Q.3.c For retrieval, each image is described with 1000 interest points, each interest point is 64-dimensional. How many computations are required when finding matches between 2 images? Now, assume comparing 1M interest points takes 0.8 seconds. How long does it take to compare an image with a dataset containing 2M images (ignoring other factors such as sorting)?

Q.3.d After retrieving results with BoW, we use geometrical verification to rerank the top 200 images. However, our evaluation shows no difference in precision and recall measured at $k = 200$. Why? How might you observe a difference in precision and recall?

4 Exercise 4: Image Classification & Object Detection

Q.4.a What is the difference between image classification, image localization, and object detection?

Q.4.b For binary linear classification, assume Bag of Words (BoW) is your feature extractor and Support Vector Machine (SVM) is your classifier. How is a "dog" vs "non-dog" classifier trained?

Q.4.c In an object detection task, suppose there are 5 classes and 9 possible locations in an image. We use three aspect ratios (1:1, 3:4, 4:3) and four different scales (0.5, 1, 1.5, 2). How many box evaluations are needed for this image in a multi-class object detection problem?

Q.4.d What is the key idea of Selective Search, and how does it increase variations in object detection?

Q.4.e What are the benefits of using Selective Search in object detection pipelines?

5 Exercise 5: Sliding Window Approach

Q.5.a What is the basic pipeline for window-based object detection? What are the weaknesses of window-based object detection?

Q.5.b For an image of size 512×512, how many windows are required to detect objects for 16 different orientations and 4 scales?

Q.5.c Assume a non-linear SVM on your computer requires about 0.02 seconds to classify each window. How many hours does it take to detect objects in $6,000$ images? How can you reduce the number of bounding boxes for detection?

Q.5.d Consider the application of a sliding window strategy in convolutional neural networks for a five-class object detection task on a 28×28 image. Assume the sliding window size is 14×14 with a stride of 2. No padding is used, meaning the convolutional kernels cannot exceed the image. How many outcomes are generated after processing by the sliding window?

Q.5.e In the case of a five-class object detection task on a 28×28 image, suppose the sliding window size is 14×14 and the stride is 2. Unlike in the previous, where no padding was used, now assume a padding size of 2. How many outcomes are generated after processing by the sliding window?